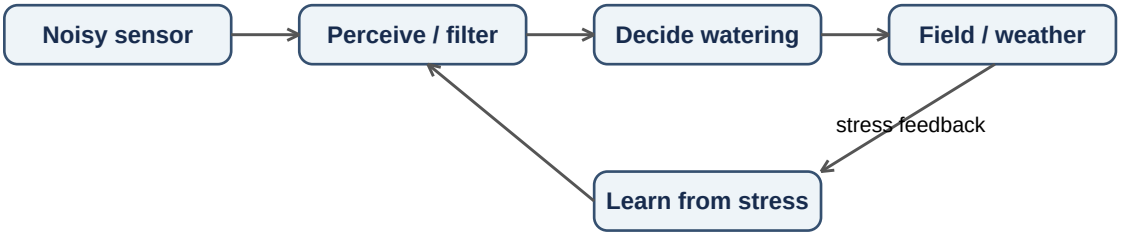
Smart Irrigation System: Case Study and Evaluation

Practical Sheet 1 | Intelligent Systems

1. Problem. A farm near Assiut wants to replace a fixed irrigation timer with an intelligent controller. The main goal is to reduce water use without increasing crop stress. The task is challenging because soil-moisture readings contain noise and the weather forecast is correct only 80% of the time. The controller must therefore make decisions under uncertainty instead of blindly following a fixed schedule.

2. Environment. The farm is represented by a simulation that runs for 90 days. Soil moisture is a value between 0 and 1. Heat and evaporation reduce moisture, rain increases it, and irrigation adds water. A stress day is recorded when moisture falls below 0.30. These values are simplified educational assumptions; they are not universal thresholds for real crops.

3. Agent and architecture. The intelligent agent receives sensor information, estimates the current moisture level, chooses an irrigation amount, and applies that action to the simulated field. It then uses the observed outcome to adjust its moisture target. The environment and agent form a feedback loop: actions change the field, and the field's next state affects the next decision.



4. Perception. The sensor reading is simulated moisture plus Gaussian noise. To avoid reacting too strongly to one unreliable measurement, the agent uses an exponential moving average: 70% of the previous estimate and 30% of the newest reading. This smooths sudden noisy changes while still allowing the estimate to respond to new information.

5. Knowledge. The agent receives a rain forecast with 80% accuracy. When rain is predicted, it reduces routine irrigation because rainfall may provide some of the required water. However, a forecast is not treated as a guarantee. If the estimated moisture becomes critically low, the controller can still provide emergency irrigation.

6. Decision. The controller compares estimated moisture with a target level and calculates how much water is needed to close the gap. A daily limit prevents excessive watering. The forecast reduces normal watering, while the low-moisture safeguard helps protect the crop. The baseline controller does not use sensor or forecast information; it applies 0.13 water units every other day.

Evaluation, Results, and Limitations

7. Action. The selected irrigation amount is applied to the simulated field. The field then changes as irrigation, rain, and heat affect its moisture. Moisture is constrained to the interval [0, 1], preventing impossible values below zero or above full saturation.

8. Learning. The agent adapts its target using feedback from the previous day. After a stress day, the target rises by 0.025, up to a maximum of 0.78. After a safe day, it may decrease slightly, but it does not fall below 0.60. This is a simple adaptive rule rather than a trained machine-learning model.

9. Evaluation method. Both controllers are evaluated for 90 simulated days using the same 20 random seeds, numbered 0 through 19. The report compares average water use, crop-stress days, and pump-on days. Using the same seeds makes the comparison fairer because both approaches experience corresponding simulated conditions. The values below are simulation outputs, not measurements from a real farm.

Metric (mean over 20 runs)	Smart agent	Fixed timer
Water units	1.67	5.85
Crop-stress days	0.00	0.00
Pump-on days	37.65	45.00

10. Effectiveness and efficiency. According to the reported simulation results, the smart agent used about 71.5% less water than the fixed timer. Both approaches recorded zero average crop-stress days, while the smart agent operated the pump on fewer days. This suggests that sensor-based decisions and forecast information can improve water efficiency in this particular model.

11. Robustness. Results depend on the assumptions and parameter values used in the simulation. Different heat levels, rain frequency, sensor noise, or stress thresholds could change the outcome. Repeated runs help reduce dependence on one random scenario, and automated tests check expected program behavior. However, these tests cannot guarantee safe performance in every real-world condition.

12. Limitations and ethics. The model represents the field with one moisture value, simplified weather, abstract water units, and a simplified stress threshold. It does not model crop species, soil type, root depth, crop yield, hardware costs, or live weather services. Before real deployment, the system would require sensor calibration, field trials, crop-specific thresholds, sensor-failure safeguards, and human oversight. It should never withhold emergency water solely because rain is forecast. Water savings must be considered together with crop health and yield.

Reproducibility. Run `python irrigation.py` to repeat the 20-seed evaluation. Run `python -m unittest discover -s tests -v` to execute the automated tests.